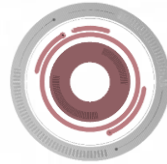




Humanoid Robot

머리 설계 사례



Humanoid Robot

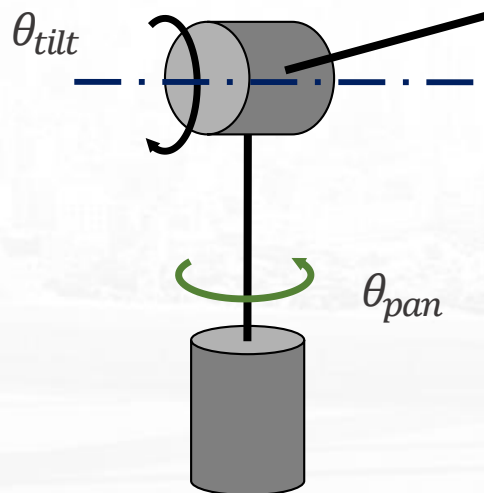
머리 및 손 설계 (Head and Hand Design)

머리 설계 사례



머리 설계 사례 : HUBO

- The head is composed of the neck and eyes.
- The Pan & Tilt mechanism is adopted for a neck and eyes.

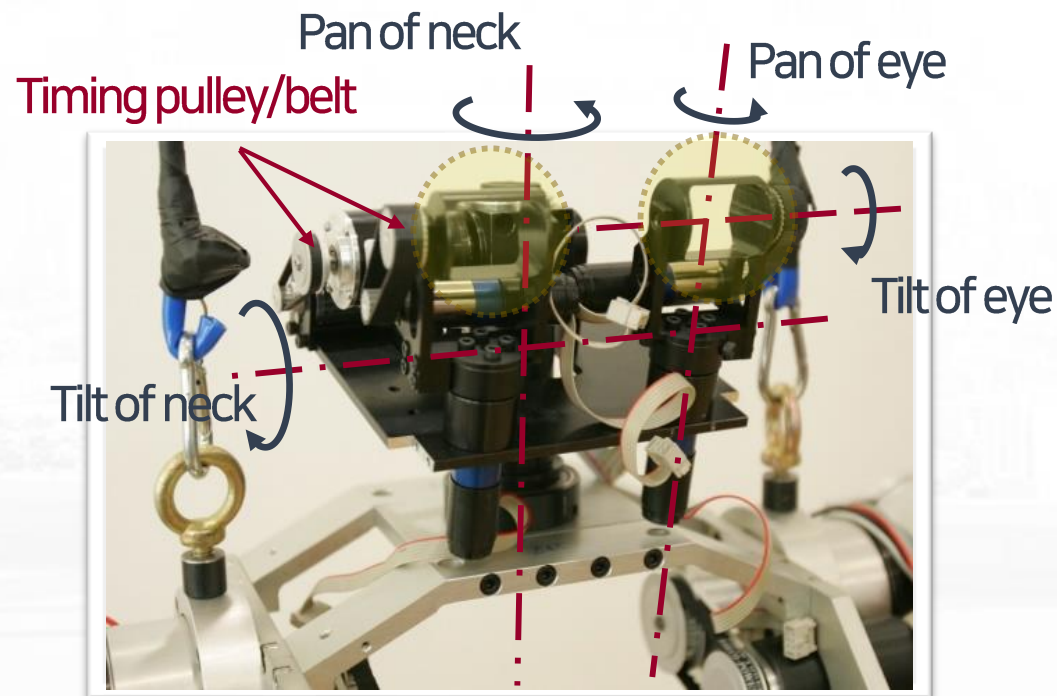


Pan & Tilt Mechanism



머리 설계 사례 : HUBO

- The head is composed of the neck and eyes.
- The Pan & Tilt mechanism is adopted for a neck and eyes.
- Total 6 DOF (2 for each eye and 2 for neck)
- Timing pulley/belt mechanisms are additionally used for tilt joints.
- CCD cameras are attached at the eye mechanism.





머리 설계 사례

Actuator specifications(Eye)

Motor



Faulhaber 1724 with gearhead





DC-Micromotors

4,2 mNm

Precious Metal Commutation

For combination with (overview on page 14-15)

Gearheads:
15A, 16A, 16/7Encoders:
IE2 – 16 ... 512

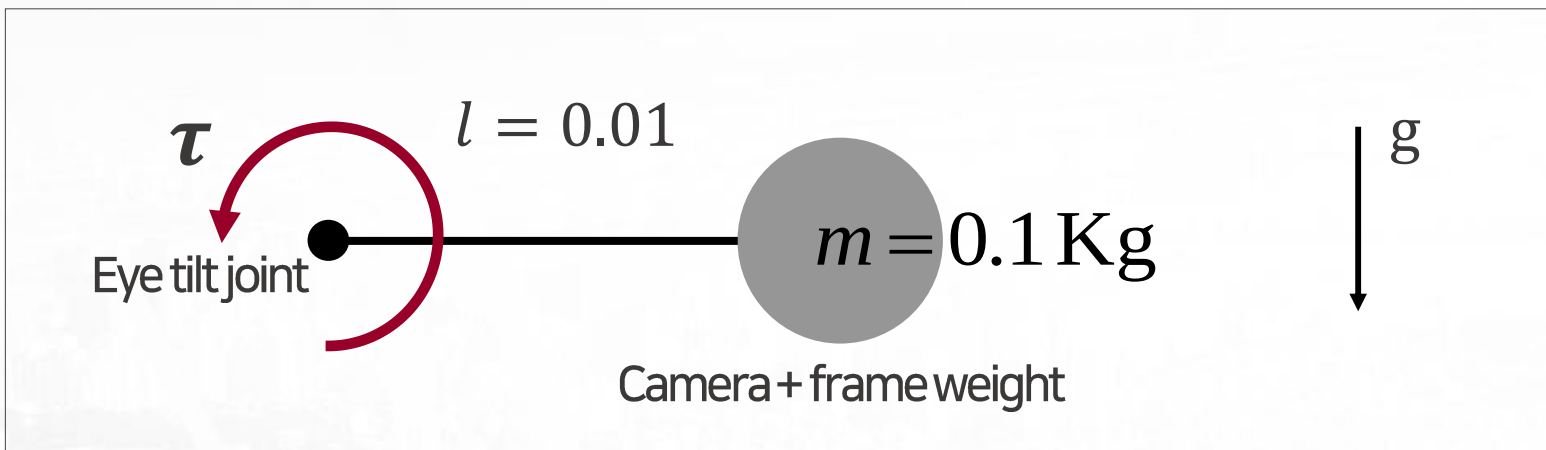
Series 1724 ... SR

	1724 T	003 SR	006 SR	012 SR	018 SR	024 SR	
1 Nominal voltage	U_N	3	6	12	18	24	Volt
2 Terminal resistance	R	0,78	3,41	16,20	32,10	54,60	Ω
3 Output power	$P_2 \text{ max.}$	2,83	2,58	2,17	2,47	2,58	W
4 Efficiency	$\eta \text{ max.}$	82	81	80	81	81	%
5 No-load speed	n_0	8 200	8 600	7 900	8 400	8 600	rpm
6 No-load current (with shaft $\varnothing$ 1,5 mm)	I_0	0,038	0,020	0,009	0,006	0,005	A
7 Stall torque	M_H	13,2	11,5	10,5	11,2	11,5	mNm
8 Friction torque	M_R	0,13	0,13	0,13	0,12	0,13	mNm
9 Speed constant	k_n	2 760	1 450	666	472	362	rpm/V
10 Back-EMF constant	k_E	0,362	0,690	1,500	2,120	2,760	mV/rpm
11 Torque constant	k_M	3,46	6,59	14,30	20,20	26,30	mNm/A
12 Current constant	k_I	0,289	0,152	0,070	0,049	0,038	A/mNm
25 Weight		27					g
26 Direction of rotation		clockwise, viewed from the front face					
27 Speed up to	$n_{\text{re max.}}$	8 000	8 000	8 000	8 000	8 000	rpm
28 Torque up to	$M_{\text{re max.}}$	4,2	4,2	4,2	4,2	4,2	mNm
29 Current up to (thermal limits)	$I_{\text{e max.}}$	1,60	0,76	0,35	0,25	0,19	A

Max. output torque of motor: $26.3 \times 0.19 \times 3 = 15 \text{ mNm}$



더 폴하버(Faulhaber) 1724 DC 모터



Required max. torque : $0.01 \times 0.1 \times 9.8 \times 1.2 = 12 \text{ mNm}$



DC-Micromotors

4,2 mNm

Precious Metal Commutation

For combination with (overview on page 14-15)

Gearheads:
15A, 16A, 16/7Encoders:
IE2 – 16 ... 512

Series 1724 ... SR

	1724 T	003 SR	006 SR	012 SR	018 SR	024 SR	
1 Nominal voltage	U_N	3	6	12	18	24	Volt
2 Terminal resistance	R	0,78	3,41	16,20	32,10	54,60	Ω
3 Output power	$P_2 \text{ max.}$	2,83	2,58	2,17	2,47	2,58	W
4 Efficiency	$\eta \text{ max.}$	82	81	80	81	81	%
5 No-load speed	n_0	8 200	8 600	7 900	8 400	8 600	rpm
6 No-load current (with shaft $\varnothing$ 1,5 mm)	I_0	0,038	0,020	0,009	0,006	0,005	A
7 Stall torque	M_H	13,2	11,5	10,5	11,2	11,5	mNm
8 Friction torque	M_R	0,13	0,13	0,13	0,12	0,13	mNm
9 Speed constant	k_n	2 760	1 450	666	472	362	rpm/V
10 Back-EMF constant	k_E	0,362	0,690	1,500	2,120	2,760	mV/rpm
11 Torque constant	k_M	3,46	6,59	14,30	20,20	26,30	mNm/A
12 Current constant	k_I	0,289	0,152	0,070	0,049	0,038	A/mNm
25 Weight		27					g
26 Direction of rotation		clockwise, viewed from the front face					
27 Speed up to	$n_{\text{pe max.}}$	8 000	8 000	8 000	8 000	8 000	rpm
28 Torque up to	$M_{\text{pe max.}}$	4,2	4,2	4,2	4,2	4,2	mNm
29 Current up to (thermal limits)	$I_{\text{e max.}}$	1,60	0,76	0,35	0,25	0,19	A

Max. output torque of motor: $26.3 \times 0.19 \times 3 = 15 \text{ mNm}$



Gearhead Selection

Metric		English					
Series	PDF	Diameter (mm)	Length (mm)	Commutation	No-load speed up to (rpm)	Stall Torque up to (mNm)	Works with
0615...S		6	15	Precious Metal	20,200	0.24	
0816...S		8	16	Precious Metal	16,500	0.41	
1016...G		10	16	Precious Metal	18,400	0.87	
1024...S		10	24	Precious Metal	14,700	2.89	
1219...G		12	19	Precious Metal	16,200	1.19	
1224...S		12	24	Precious Metal	13,100	3.69	
1224...SR		12	24	Precious Metal	13,800	5.43	
1319...SR		13	19	Precious Metal	14,600	2.91	
1331...SR		13	31	Precious Metal	10,600	11.2	
1336...C		13	36	Graphite	9,200	8.4	
1516...S		15	16	Precious Metal	16,200	1.04	
1516...SR		15	16	Precious Metal	12,900	1.61	
1524...SR		15	24	Precious Metal	10,800	7.12	
1624...S		16	24	Precious Metal	14,400	5.16	
1717...SR		17	17	Precious Metal	14,000	5.38	
1724...SR		17	24	Precious Metal	8,600	13.2	



Gearhead Selection

1724...SR		17	24	Precious Metal	8,600	13.2	-
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Product Combinations for the 1724...SR.

Drive Electronics Controls for DC Motors	Encoders Integrated Encoders	Precision Gearheads Planetary Gearheads
» MCDC 3003/06 C	» IE2-16	» 15A
» MCDC 3003/06 S	» IE2-512	» 16/7
» MCDC 5004	Magnetic Encoders (North America)	Spur Gearheads
	» HEM2-16	» 16A

All plastic : large backlash, light

Metal : less backlash than plastic

Metal & plastic : large backlash



머리 설계 사례

Planetary Gearheads

Series 16/7



For combination with (overview on page 14-15)

DC-Micromotors:

1516, 1524, 1624, 1717, 1724, 1727

Brushless DC-Servomotors:

1524, 1536, 1628

DC-Motor-Tacho Combinations:

1841

Housing material

Geartrain material

Recommended max. input sp

- continuous operation

Backlash, at no-load

Bearings on output shaft

Shaft load, max.:

- radial (6,5 mm from mounting face)

- axial

Shaft press fit force, max.

Shaft play (on bearing output):

- radial

- axial

Operating temperature range

16/7

metal

all steel

5 000 rpm

 ω 1°

- preloaded ball bearings

 $\leq 30 \text{ N}$ $\leq 5 \text{ N}$ $\leq 5 \text{ N}$ $\leq 0,02 \text{ mm}$
$$= 0 \text{ mm}$$

– 30 ... + 100 °C

Specifications

reduction ratio (nominal)	weight without motor	length without motor L2 mm	length with motor						output torque		direction of rotation (reversible)	efficiency %
			1516 T	1524 T 1624 T	1628 T	1717 T	1724 T	1727 U	continuous operation	intermittent operation		
			L1 mm	L1 mm	L1 mm	L1 mm	L1 mm	L1 mm	M max. mNm	M max. mNm		
3,71:1	18	17,0	32,8	40,8	45,0	34,0	41,0	44,2	200	300	=	90
14 :1	23	21,2	36,9	44,9	49,1	38,1	45,1	48,3	300	450	=	80
43 :1	28	25,3	41,1	49,1	53,3	42,3	49,3	52,5	300	450	=	70
66 :1	28	25,3	41,1	49,1	53,3	42,3	49,3	52,5	300	450	=	70
134 :1	33	29,4	45,2	53,2	57,4	46,4	53,4	56,6	300	450	=	60
159 :1	33	29,4	45,2	53,2	57,4	46,4	53,4	56,6	300	450	=	60
246 :1	33	29,4	45,2	53,2	57,4	46,4	53,4	56,6	300	450	=	60



「플레네터리 기어헤드(planetary gearhead)

Max. permissible torque of gearhead
: 450 mNm > 12 mNm

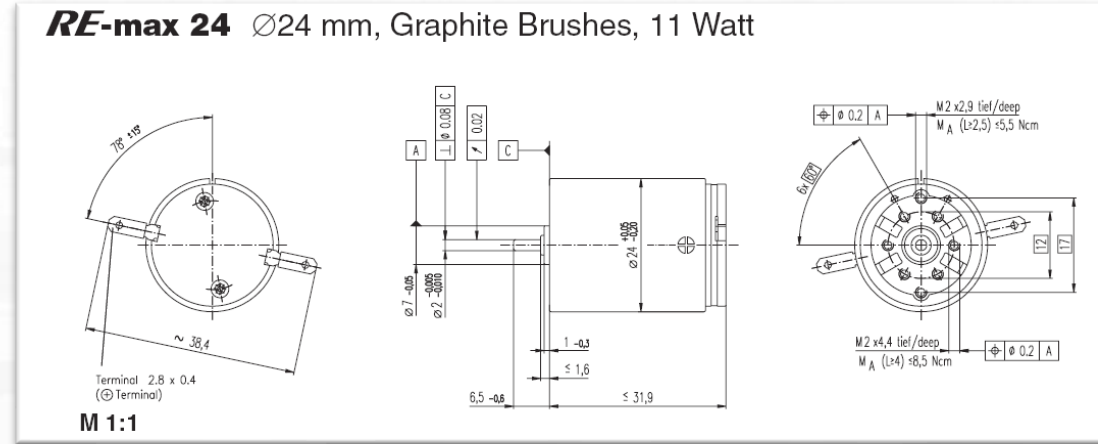
Max. output torque of geared motor :
 $15 \text{ mNm} \times 246 \times 1.56 \times 0.6 = 3,454 \text{ mNm} > 12 \text{ mNm}$

Actuator specifications(Neck)

Head	Neck	Pan	Planetary gear (109:1)	N/A	RE-Max 24 (11W)
		Tilt	Miniature harmonic gear (100:1)	Timing pulley/belt (2:1)	RE-Max 24 (11W)
	Eye	Pan	Planetary gear (246:1)	N/A	Faulhaber 1724 (3.7W)
		Tilt	Planetary gear (246:1)	Timing pulley/belt (1.56:1)	Faulhaber 1724 (3.7W)



RE-Max 24



Dimension of RE-Max 24 DC motor



- ☒ Stock program
☐ Standard program
☐ Special program (on request)

Order Number

222036	222037	222038	222039	222040	222041	222042	222043	222044	222045	222046	222047					
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Motor Data

Values at nominal voltage

1	Nominal voltage	V	9.0	12.0	15.0	18.0	18.0	24.0	24.0	36.0	42.0	48.0	48.0	48.0			
2	No load speed	rpm	8190	7540	8290	8870	8000	9310	7540	9120	8600	7410	5250	4730			
3	No load current	mA	45.6	31.0	27.7	25.0	22.1	19.8	15.5	12.9	10.3	7.60	5.15	4.59			
4	Nominal speed	rpm	6640	5870	6650	7240	6330	7670	5850	7450	6920	5690	3470	2960			
5	Nominal torque (max. continuous torque)	mNm	10.8	12.4	12.4	12.3	12.3	12.1	12.2	12.1	12.0	12.0	12.0	12.1			
6	Nominal current (max. continuous current)	A	1.08	0.853	0.752	0.666	0.598	0.516	0.421	0.335	0.270	0.204	0.144	0.131			
7	Stall torque	mNm	57.7	56.9	63.6	67.7	59.5	69.3	55.3	66.6	62.0	52.4	35.7	32.8			
8	Starting current	A	5.55	3.78	3.71	3.52	2.79	2.83	1.83	1.78	1.34	0.855	0.414	0.343			
9	Max. efficiency	%	83	83	83	84	83	84	83	84	83	82	79	78			

Characteristics

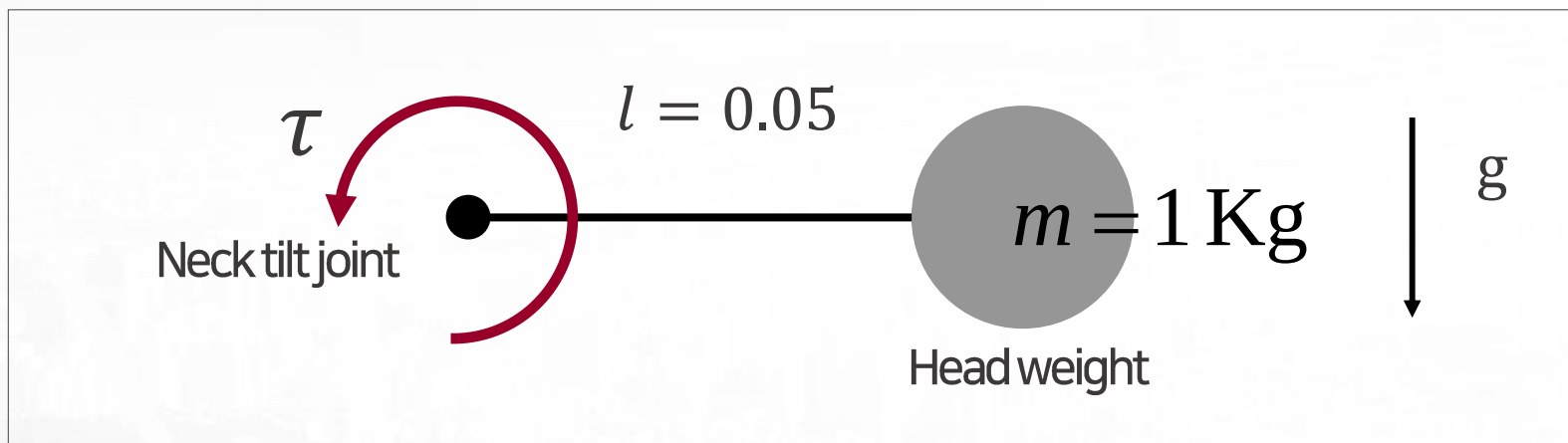
10	Terminal resistance	Ω	1.62	3.18	4.05	5.11	6.44	8.47	13.1	13.1	13.1	13.1	13.1	13.1			
11	Terminal inductance	mH	0.0735	0.154	0.200	0.251	0.309	0.406	0.618	0.618	0.618	0.618	0.618	0.618			
12	Torque constant	mNm / A	10.4	15.1	17.2	19.2	21.3	24.4	30.1	30.1	30.1	30.1	30.1	30.1			
13	Speed constant	rpm / V	919	634	557	497	448	391	317	317	317	317	317	317			
14	Speed / torque gradient	rpm / mNm	143	134	131	132	135	135	138	138	138	138	138	138			
15	Mechanical time constant	ms	5.96	5.90	5.88	5.89	5.91	5.92	5.92	5.92	5.92	5.92	5.92	5.92			
16	Rotor inertia	gcm ²	3.97	4.22	4.28	4.26	4.17	4.17	4.11	4.11	4.07	4.00	3.88	3.92			

- Nominal Voltage : 24V
- Torque Constant : 24.4 mNm/A
- Nominal Current : 0.516 A

Max. output torque of motor : $24.4 \times 0.516 \times 3 = 38 \text{ mNm}$



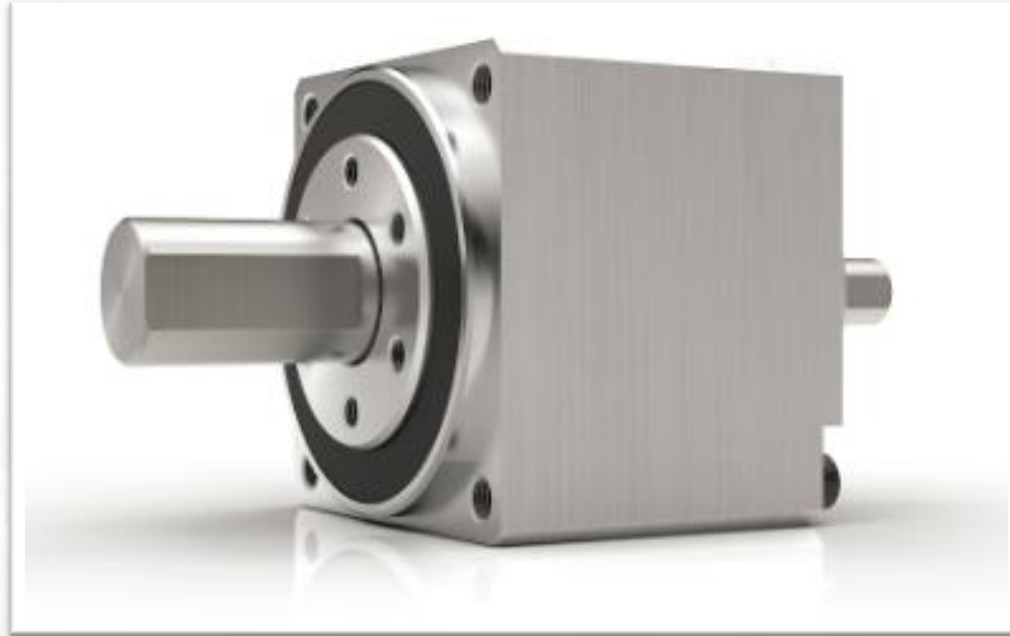
RE-max 24



Required max. torque : $0.05 \times 1 \times 9.8 \times 1.2 = 588 \text{ mNm}$



감속기 : Type 1U (Double shaft)



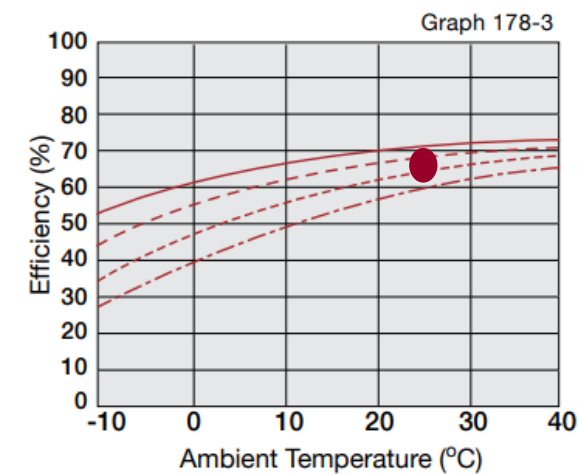


Rating Table

Size	Gear Ratio	Rated Torque at 2000rpm		Repeated peak torque		Max. average load torque		Max. momentary torque		Max. Input Speed	Avg. Input Speed	Moment of Inertia(1/4 GD) ²	
		Nm	in.lb	Nm	in.lb	Nm	in.lb	Nm	in.lb			1U	2XH
5	30	0.25	2	0.5	4	0.38	3	0.9	8	10,000	6,500	2.5×10^{-4}	2.5×10^{-4}
	50	0.4	4	0.9	8	0.53	5	1.8	16				
	100	0.6	5	1.4	12	0.94	8	2.7	24				
8	30	0.9	8	1.8	16	1.4	12	3.3	29	8,500	3,500	3.0×10^{-3}	3.2×10^{-3}
	50	1.8	16	3.3	29	2.3	20	6.6	58				
	100	2.4	21	4.8	42	3.3	29	9.0	80				
11	30	2.2	19	4.5	40	3.4	30	8.5	75	8,500	3,500	1.2×10^{-2}	1.4×10^{-2}
	50	3.5	31	8.3	73	5.5	49	17	150				
	100	5.0	44	11	97	8.9	79	25	221				
14	30	4.0	35	9.0	80	6.8	60	17	150	8,500	3,500	3.3×10^{-2}	3.4×10^{-2}
	50	5.4	48	18	159	6.9	61	35	310				
	80	7.8	69	23	204	11	97	47	416				
	100	7.8	69	28	248	11	97	54	478				

Table 2

Ratio 100

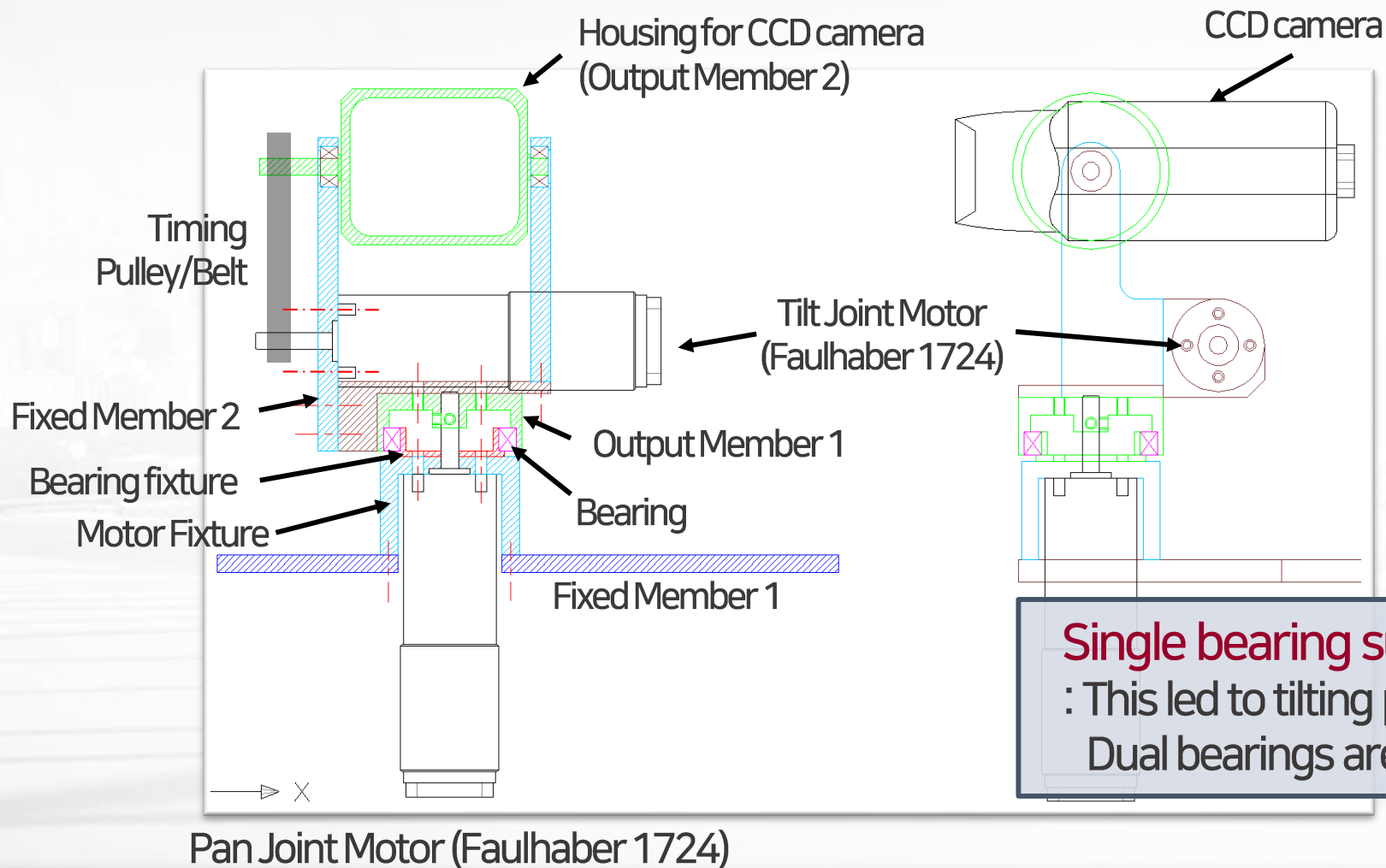


Max. permissible torque of gearhead : $2,700 \text{ mNm} > 588 \text{ mNm}$

Max. output torque of actuator : $38 \text{ mNm} \times 100 \times 2 \text{ (timing pulley ratio)} \times 0.65 \text{ (efficiency)} = 4,940 \text{ mNm} > 588 \text{ mNm}$



Eyes Pan & Tilt Joints





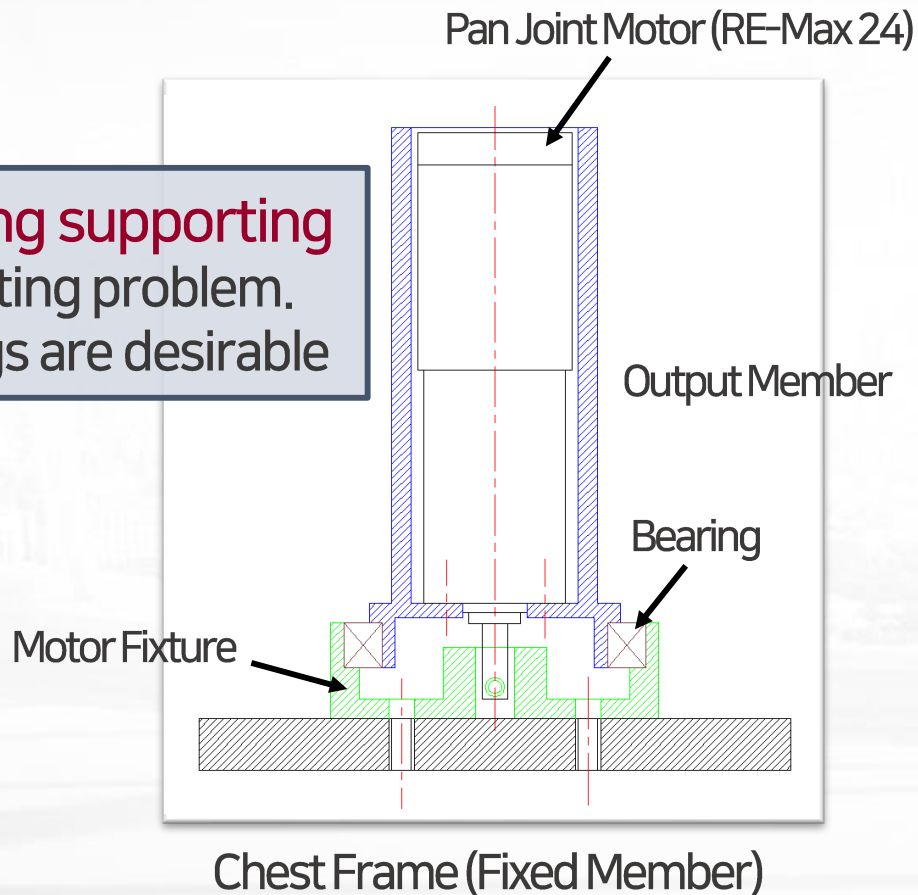
Neck Pan Joint

- Direct drive without timing pulley/belt mechanism
- Planetary gearhead (109:1)



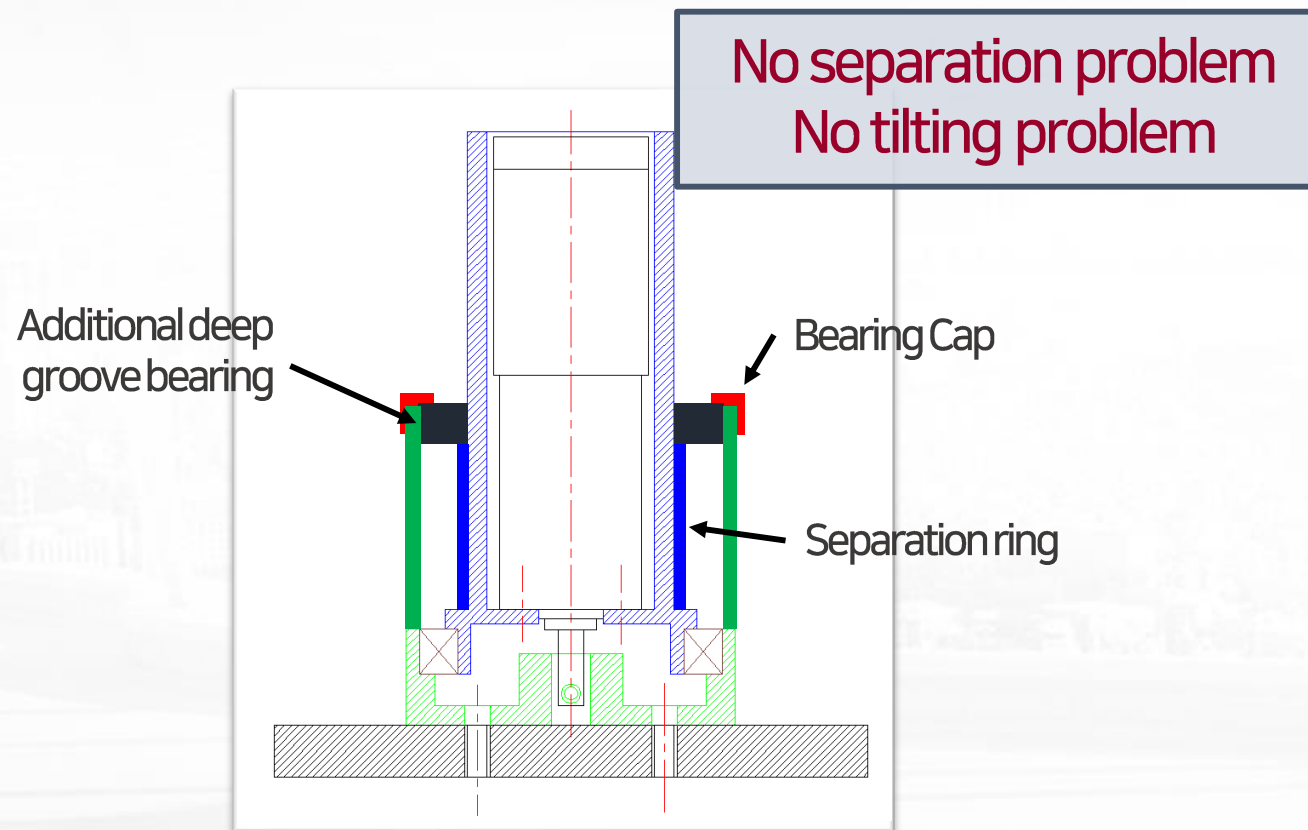
Neck Pan Joint

Single bearing supporting
: This led to tilting problem.
Dual bearings are desirable





Neck Pan Joint- Improved Design



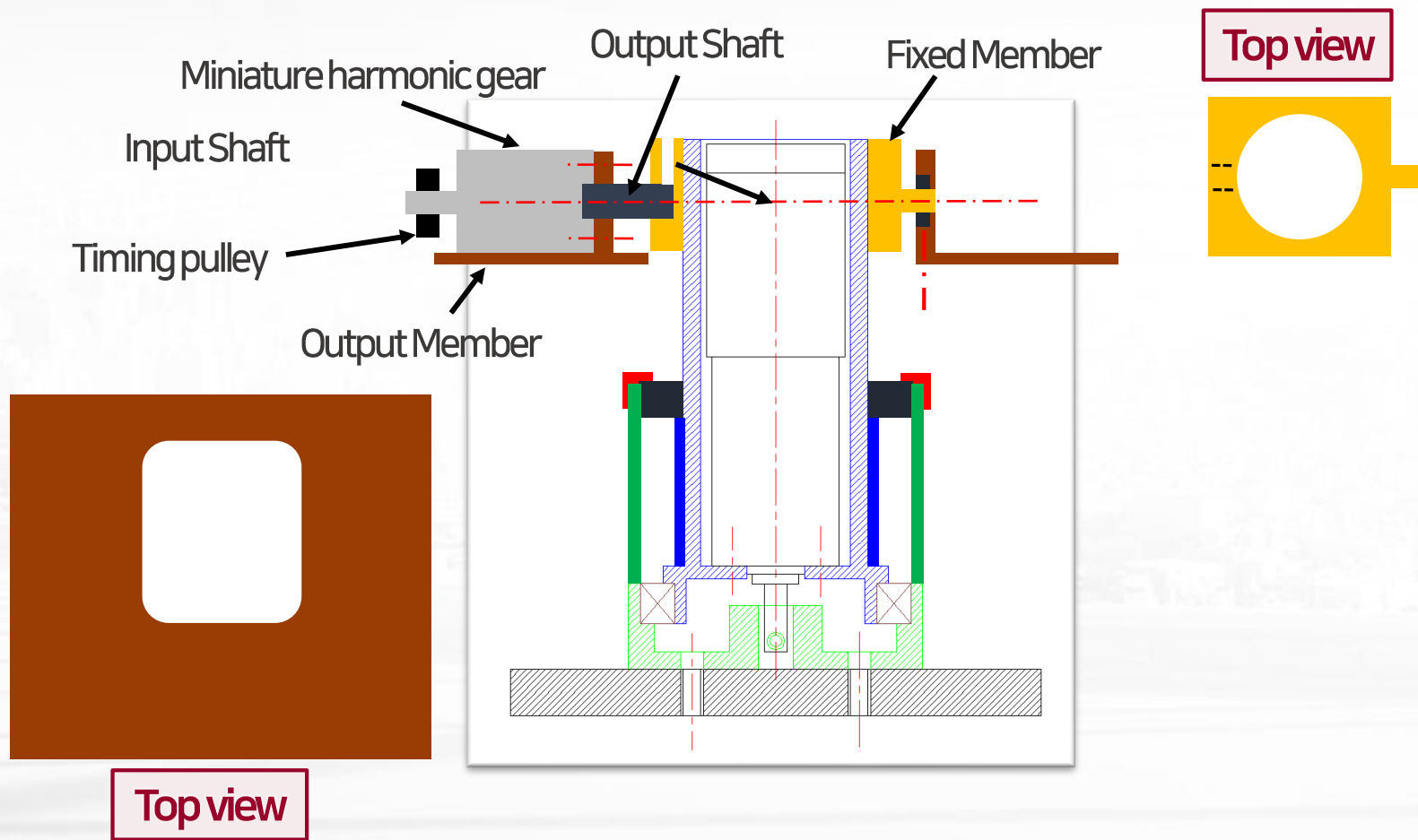


Neck Pan Joint

- ▶ Indirect drive using a timing pulley/belt mechanism
- ▶ Miniature harmonic gear (100:1)

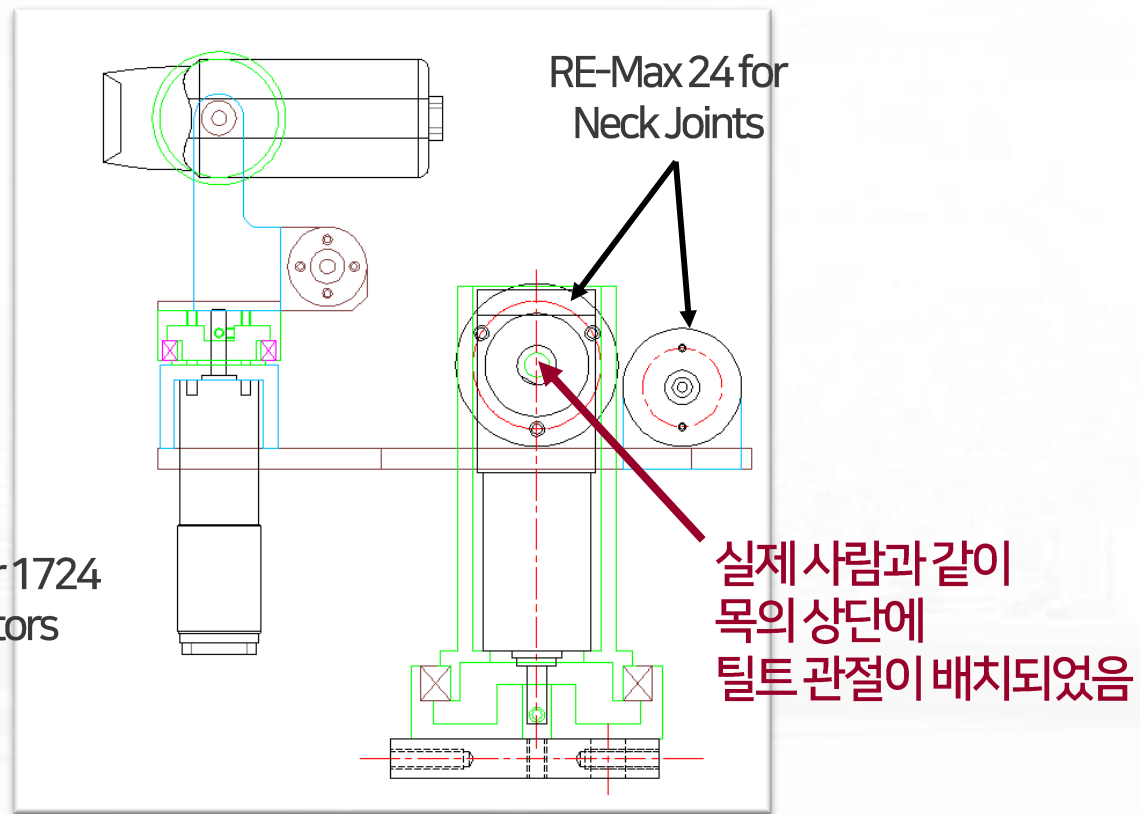
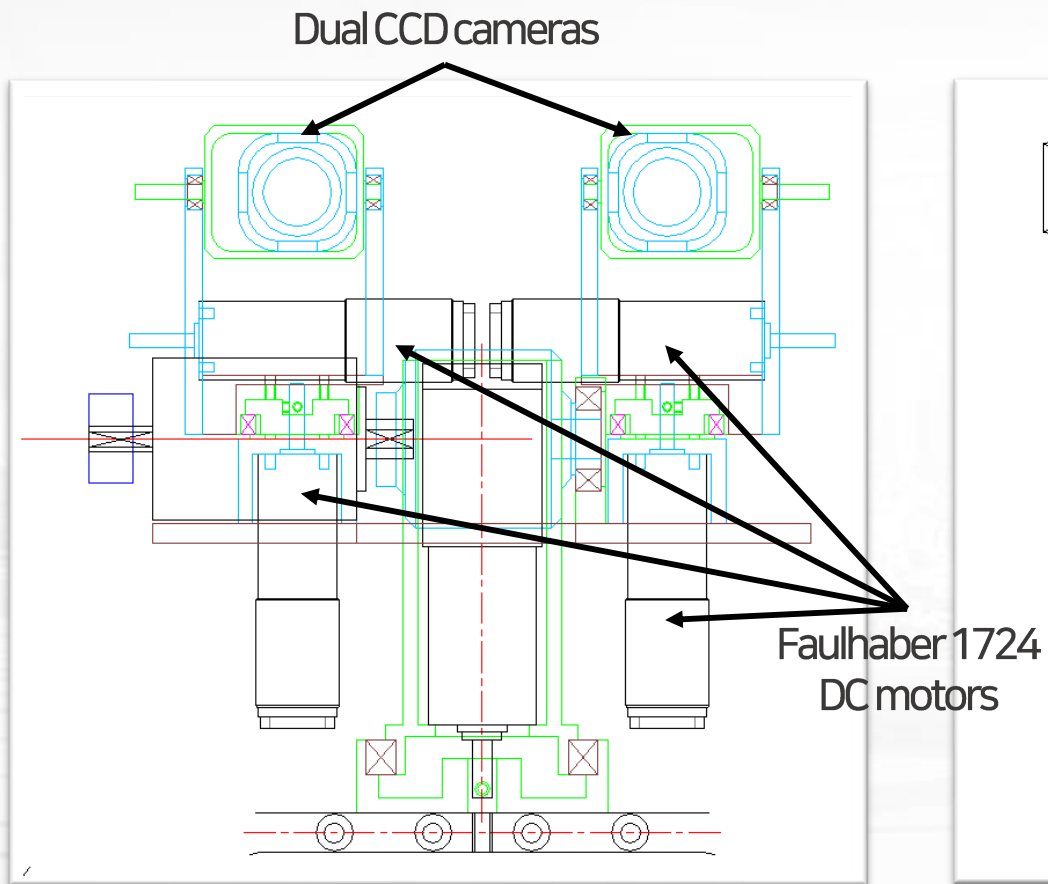


Neck Pan Joint





2D Assembly Drawing





「Motion Test of Hand & Head



Open-loop motion test



「Motion Test of Hand & Head



Closed-loop motion test
(Gaze control)